2 Para evaluar las integrales (a) $\int \operatorname{sen} mx \cos nx\, dx$, (b) $\int \operatorname{sen} mx \operatorname{sen} nx\, dx$ o (c) $\int \cos mx \cos nx\, dx$, utilice la identidad correspondiente:

$$\text{(a) } \operatorname{sen} A \cos B = \tfrac{1}{2}[\operatorname{sen}(A - B) + \operatorname{sen}(A + B)]$$

$$\text{(b) } \operatorname{sen} A \operatorname{sen} B = \tfrac{1}{2}[\cos(A - B) - \cos(A + B)]$$

$$\text{(c) } \cos A \cos B = \tfrac{1}{2}[\cos(A - B) + \cos(A + B)]$$

7.2 EJERCICIOS

1–49 Evalúe la integral.

1. $\int \operatorname{sen}^3 x \cos^2 x\, dx$

2. $\int \operatorname{sen}^6 x \cos^3 x\, dx$

5. $\int \operatorname{sen}^5(2t) \cos^2(2t)\, dt$

6. $\int t \cos^5(t^2)\, dt$

8. $\int \dfrac{\operatorname{sen}^3(\sqrt{x})}{\sqrt{x}}\, dx$

10. $\int x \operatorname{sen}^3 x\, dx$

13. $\int \sqrt{\cos\theta}\, \operatorname{sen}^3\theta\, d\theta$

14. $\int \dfrac{\operatorname{sen}^2(1/t)}{t^2}\, dt$

15. $\int \cot x \cos^2 x\, dx$

16. $\int \tan^2 x \cos^3 x\, dx$

17. $\int \operatorname{sen}^2 x \operatorname{sen} 2x\, dx$

18. $\int \operatorname{sen} x \cos\left(\tfrac{1}{2}x\right) dx$

19. $\int t \operatorname{sen}^2 t\, dt$

20. $\int \cos\theta \cos^5(\operatorname{sen}\theta)\, d\theta$

21. $\int \tan x \sec^3 x\, dx$

22. $\int \tan^2\theta \sec^4\theta\, d\theta$

23. $\int \tan^2 x\, dx$

24. $\int (\tan^2 x + \tan^4 x)\, dx$

25. $\int \tan^4 x \sec^6 x\, dx$

26. $\int_0^{\pi/4} \sec^6\theta \tan^6\theta\, d\theta$

27. $\int \tan^3 x \sec x\, dx$

28. $\int \tan^5 x \sec^3 x\, dx$

29. $\int \tan^3 x \sec^6 x\, dx$

31. $\int \tan^5 x\, dx$

32. $\int \tan^2 x \sec x\, dx$

33. $\int x \sec x \tan x\, dx$

34. $\int \dfrac{\operatorname{sen} \phi}{\cos^3 \phi}\, d\phi$

36. $\int \csc^4 x \cot^6 x\, dx$

39. $\int \csc x\, dx$

41. $\int \operatorname{sen} 8x \cos 5x\, dx$

42. $\int \operatorname{sen} 2\theta \operatorname{sen} 6\theta\, d\theta$

43. $\int_0^{\pi/2} \cos 5t \cos 10t\, dt$

44. $\int \operatorname{sen} x \sec^5 x\, dx$

46. $\int \dfrac{\cos x + \operatorname{sen} x}{\operatorname{sen} 2x}\, dx$

47. $\int \dfrac{1 - \tan^2 x}{\sec^2 x}\, dx$

48. $\int \dfrac{dx}{\cos x - 1}$

49. $\int x \tan^2 x\, dx$

51–54 Evalúe las integrales indefinidas siguientes. Ilustre y verifique que su respuesta es razonable, al trazar la gráfica del integrando y su antiderivada (tome $C = 0$).

51. $\int x \operatorname{sen}^2(x^2)\, dx$

52. $\int \operatorname{sen}^5 x \cos^3 x\, dx$

53. $\int \operatorname{sen} 3x \operatorname{sen} 6x\, dx$

54. $\int \sec^4\left(\frac{1}{2}x\right) dx$
